

NEWTON 11 SW, GA, USA

WMO: 723220

Lat: 31.1923N

Lon: 84.4465W

Elev: 156

StdP: 14.61

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 63829

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.5	29.6	13.1	10.8	37.5	17.8	13.6	40.0	10.2	54.7	8.9	52.1	1.3	N/A	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	94.8	74.9	92.9	75.3	91.2	75.1	79.3	88.6	78.4	87.4	77.7	86.5	4.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.1	142.0	82.7	76.2	137.7	82.0	75.4	133.8	81.2	42.8	88.9	41.9	87.9	41.2	86.6	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.6	7.5	6.7	20.7	96.9	3.1	2.6	18.5	98.8	16.6	100.3	14.9	101.8	12.6	103.7	
			19.5	80.8	3.7	1.3	16.8	81.7	14.7	82.5	12.6	83.2	9.9	84.1	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.8	49.9	53.5	59.8	66.1	73.9	79.4	80.9	80.6	77.1	68.1	57.9	53.0
	DBStd	13.02	9.98	8.79	8.04	6.12	5.08	3.05	2.46	2.60	4.06	7.17	8.21	9.21
	HDD50	303	130	60	16	1	0	0	0	0	0	1	22	73
	HDD65	1751	475	331	199	59	7	0	0	0	1	53	242	384
	CDD50	6417	126	157	319	485	740	882	957	950	812	562	259	168
	CDD65	2389	6	9	36	93	282	432	492	485	363	149	29	13
	CDH74	21612	19	63	313	910	2663	4095	4634	4359	3088	1223	199	46
	CDH80	8840	0	3	40	196	1039	1874	2124	1923	1279	339	23	0
Wind	WSAvg	2.7	2.9	3.1	3.2	3.1	2.8	2.5	2.2	2.1	2.5	2.5	2.6	2.7
Precipitation	PrecAvg	49.6	4.8	4.0	5.6	3.1	2.3	5.4	6.0	4.8	4.7	2.8	3.1	3.5
	PrecMax	69.0	14.9	6.7	11.2	7.4	6.1	9.8	9.8	9.3	14.4	8.6	9.0	6.5
	PrecMin	32.7	0.9	0.6	0.4	0.6	0.1	1.5	3.6	2.2	0.8	0.4	0.4	0.9
	PrecStd	9.4	3.0	1.9	3.5	1.7	1.5	2.6	2.0	2.0	3.4	1.9	2.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	79.9	83.8	87.3	93.7	97.5	95.8	96.4	95.4	89.6	83.0	78.1
		MCWB	65.5	67.1	64.9	69.3	71.3	72.4	76.6	75.4	73.8	72.3	69.1	68.4
	2%	DB	73.1	75.8	80.8	84.8	90.6	94.8	93.5	93.4	91.7	86.4	79.3	75.1
		MCWB	64.0	64.0	64.9	68.0	70.5	74.0	76.4	76.5	73.6	69.2	66.6	66.0
	5%	DB	69.7	71.9	77.6	82.3	88.4	92.2	92.1	91.5	89.4	84.0	76.0	71.6
		MCWB	62.8	63.5	64.0	66.5	70.1	75.0	76.4	76.3	73.6	67.6	64.4	65.7
	10%	DB	66.1	68.6	74.6	79.7	86.1	89.5	90.2	89.5	87.4	81.2	72.7	68.5
		MCWB	60.7	61.7	62.1	64.9	69.6	74.5	76.4	76.5	73.3	67.2	63.5	64.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.7	69.5	69.6	73.0	75.5	79.3	80.2	80.8	78.5	77.0	73.6	71.6
		MCDB	72.5	75.0	76.2	81.1	84.4	89.2	89.4	90.5	87.1	83.4	78.1	75.3
	2%	WB	66.5	68.0	68.2	71.4	74.4	78.1	79.2	79.5	77.3	74.9	71.2	69.6
		MCDB	70.5	73.9	74.7	79.1	83.0	87.7	89.3	88.1	85.9	80.3	75.2	72.7
	5%	WB	64.4	66.0	66.6	70.0	73.4	77.2	78.3	78.5	76.4	72.9	68.8	67.9
		MCDB	68.5	71.6	73.5	77.8	81.6	86.5	87.9	86.9	85.0	78.9	73.0	71.3
	10%	WB	62.0	63.7	65.0	68.4	72.4	76.3	77.6	77.8	75.4	71.0	65.8	65.5
		MCDB	66.3	68.8	72.4	75.9	81.1	85.0	86.8	86.0	83.9	77.9	71.5	68.9
Mean Daily Temperature Range	5% DB	MDBR	21.4	21.8	22.8	23.1	22.0	19.3	18.6	17.8	19.3	22.4	23.8	20.9
		MCDBR	22.2	24.5	24.9	25.7	24.6	23.7	21.0	20.8	22.4	24.6	24.8	21.3
		MCWBR	13.5	14.5	11.5	10.6	8.1	7.1	6.6	6.9	8.0	11.2	13.3	13.6
	5% WB	MCDBR	18.4	20.2	20.3	19.9	19.2	18.8	18.9	18.1	18.8	18.6	18.1	17.5
		MCWBR	13.1	14.2	11.2	10.9	7.3	7.3	6.9	6.9	7.0	9.9	11.5	12.8
Clear-Sky Solar Irradiance	taub	0.327	0.334	0.355	0.385	0.434	0.482	0.508	0.496	0.426	0.374	0.351	0.333	
	taud	2.510	2.486	2.441	2.358	2.252	2.180	2.119	2.151	2.334	2.451	2.458	2.512	
	Ebn at Noon	286	294	294	288	273	259	251	253	268	276	274	278	
	Edh at Noon	28	31	35	39	44	47	50	48	38	32	29	27	
All-Sky Solar Radiation	RadAvg	965	1146	1524	1868	2048	1923	1900	1743	1575	1370	1081	842	
	RadStd	62	135	126	148	141	143	119	121	128	134	84	80	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.53	N/A	N/A	N/A	N/A	N/A	-86	-284	+466	+238
(47)	Regional (37 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+288	+184

Nomenclature: See separate page